Late Devonian paleomagnetic age for the Polaris Mississippi Valley-type Zn–Pb deposit, Canadian Arctic Archipelago[1]

D. T. A. SYMONS
Department of Geology, University of Windsor, Windsor, Ont., Canada N9B 3P4

AND

D. F. SANGSTER
Geological Survey of Canada, 601 Booth Street, Ottawa, Ont., Canada K1A 0E8

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The Polaris Mississippi Valley-type (MVT) zinc–lead deposit is located in the Middle to Upper Ordovician Thumb Mountain Formation of the Cornwallis Fold Belt in the Canadian Arctic Archipelago. Paleomagnetic analysis was done on samples from 29 sites in the sphalerite–galena ore, the surrounding dolomitic envelope, and the adjacent host-rock limestones, using alternating-field and thermal step demagnetization, scanning electron microscopy with energy-dispersive analysis, and saturation isothermal remanence analyses. Excluding the modern viscous remanence component, the ore, dolomite alteration, and host limestones carry only a prefolding *A* remanent magnetization component that resides in single- and pseudosingle-domain magnetite, both as discrete grains and as inclusions in the MVT ore minerals. All evidence indicates that the mineralization and magnetization events were coeval. The mean magnetization direction of $D = 145.8°$, $I = -31.1°$ ($\alpha_{95} = 5.9°$, $k = 26.2$, $N = 24$) after tilt and a minor plunge correction gives a Late Devonian pole position of 28.6°N, 121.2°E (dp = 3.7°, dm = 6.6°, $N = 24$). The results indicate that the ore deposit was formed during the onset of the Ellesmerian Orogeny and tilted later in the orogeny.

Le gîte de zinc–plomb de Polaris, de type Mississippi Valley (TMV), est localisé dans la Formation de Thumb Mountain, d'âge ordovicien moyen–supérieur, dans la zone de plissements de Cornwallis de l'archipel Arctique canadien. Une étude paléomagnétique a porté sur les échantillons prélevés à 29 sites : dans le minerai sphalérite–galène, dans l'enveloppe dolomitique qui l'entoure et dans les roches calcaires encaissantes voisines, utilisant la désaimantation par champ alternatif et par chauffage, la microscopie à balayage avec analyse par énergie dispersive, et les analyses de la rémanence isotherme de saturation. Après élimination de la composante de rémanence visqueuse actuelle, il est constaté que le minerai, l'altération dolomitique et les calcaires encaissants portent seulement une composante d'aimantation rémanente *A*, acquise antérieurement au plissement, et qui réside dans les monodomaines ou pseudomonodomaines de la magnétite, les deux se trouvant aussi bien dans les grains uniques que dans les inclusions des minéraux métalliques du gîte de type MV. Toutes les observations indiquent que les événements de minéralisation et d'aimantation étaient contemporains. La direction moyenne d'aimantation de $D = 145{,}8°$, $I = -31{,}1°$ ($\alpha_{95} = 5{,}9°$, $k = 26{,}2$, $N = 24$), après correction pour basculement et faible plongée, fournit un paléopôle avec une direction de 28,6°N, 121,2°E (dp = 3,7°, dm = 6,6°, $N = 24$) qui correspond au Dévonien tardif. Les résultats indiquent que le gîte s'est formé au début de l'orogenèse ellesmérienne, et qu'il fut incliné plus tard durant l'orogenèse.

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Introduction

Nearly all Mississippi Valley-type (MVT) zinc–lead ore deposits lack crosscutting relationships, deformational events, or other geologic features to delimit their age, and they lack suitable minerals to measure their age by radiometric methods. Until their ages are known, it is not possible to choose between several competing hypotheses for their ore genesis (Anderson and Macqueen 1982; Sangster 1986). The Polaris deposit in Canada's Arctic Islands fits the MVT stereotype. Based on slim geological evidence, Kerr (1977) suggests a mineralization age of late Early to mid Late Devonian, whereas Randell and Anderson (1990) suggest mineralization took place between the Early Silurian and Early Devonian. This study uses paleomagnetic methods to show that the Polaris MVT deposit was formed in Late Devonian time.

Geology

Polaris Zn–Pb mine is located on Little Cornwallis Island in the Arctic Archipelago of Canada at 75°30′N, 96°30′W (Fig.1). It is one of at least 14 MVT mineralized zones, most of which occur in the Thumb Mountain Formation of the Cornwallis Fold Belt (Kerr 1977).

Original reserves of the Polaris deposit were 20×10^6 t of 14.3% Zn, 3.8% Pb, 5% Fe, and 3 g/t Ag. It is hosted by platform carbonates in the upper half of the Middle to Upper Ordovician Thumb Mountain Formation (Fig.2) that is part of a Cambrian to Devonian sequence of carbonates, evaporites, shales, and sandstones that cover much of the archipelago. The Thumb Mountain Formation comprises 260 m of dense limestones and dolostones overlain by 90 m of burrowed bioclastic argillaceous wackestone. It is underlain by the 440 m thick Bay Fiord Formation of interbedded limestones, dolostones, and shales with minor gypsum and anhydrite beds, and it is overlain by the 60 m thick Irene Bay green mudstone with limestone interbeds, which in turn are overlain by the Cape Phillips shales.

The deposit is about 700 m long, 300 m wide, and 30–150 m thick. The sphalerite occurs as disseminations, open space fillings, colloform masses, veins, and veinlets. It coexists with and is replaced by galena. Marcasite and sparry dolomite both predate and postdate the sphalerite and galena. Calcite with rare gypsum and barite postdates ore deposition. Internal sedi-

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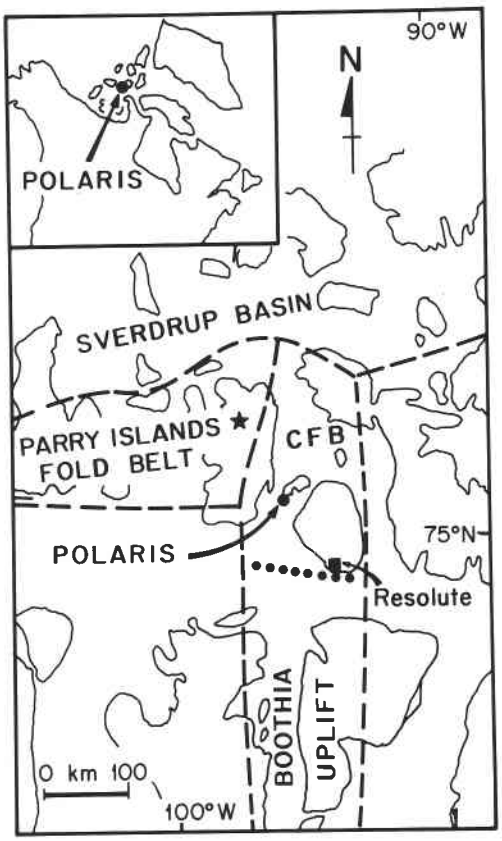


FIG. 1. Location of the Polaris mine and tectonic elements from Trettin (1989). CFB, Cornwallis Fold Belt; ★, Earth's north magnetic pole.

ments contain Upper Ordovician fossils, fragments of colloform sulphides, and mineralized rock clasts in layers that are parallel to bedding. This evidence for a pretilting age for mineralization led Randell and Anderson (1990) to suggest that ore deposition predates an Early Devonian folding event. Several marker horizons are preserved in the ore, presumably through nonreplacement by mineralization. Around the ore body the original host limestones have been altered to dolomite to form an envelope that is over 100 m thick in places. Dissolution and vein breccias occur but rarely show significant clast rotation (Randell and Anderson 1990), preventing their use in a paleomagnetic "conglomerate" test (e.g., Symons and Sangster 1991). At Polaris, the strata strike N20°W and dip 20 ± 10°E on the flank of a north-trending anticline that plunges 3°N (R. N. Randell, personal communication, 1990). A minor anticlinal fold on the west side of the property was used for a paleomagnetic fold test (Graham 1949) (Fig.2).

The Phanerozoic sedimentary rocks of the Cornwallis Fold Belt form part of the Arctic Platform. They rest on a Proterozoic basement that is exposed in the Boothia Uplift, which is the southward extension of the fold belt (Fig. 1). Uplift of up to 7–10 km in the belt is recorded in unconformities and flanking syntectonic sediments (Okulitch *et al.* 1986). Originally attributed to isostatic forces, uplift is now related to compressional deformation that produced east-dipping reverse faults and north–south-trending folds (Okultich *et al.* 1986; Trettin 1989). The folds are west verging with intervening east-facing monoclines. The intensity of folding diminishes eastward onto Cornwallis Island and increases westward into Parry Islands Fold Belt to give steeply overturned tight folds that are overthrust to the west. Kerr (1977) proposed four pulses of diastrophism, termed the Cornwallis Disturbance, in the (1) Early Silurian, (2) early Early Devonian, (3) late Early Devonian, and (4) Late Devonian, with pulse 1 being mild, pulse 2 being strong to produce most of the uplift and folding, and pulses 3 and 4 being moderately intense to reinforce the folds initiated in pulse 2. Okulitch *et al.* (1986) suggested that pulses 2 and 3 were essentially one continuous pulse in the Early Devonian which produced uplift by the stacking of thrust sheets on east-dipping basement thrust faults. Miall (1986) supported the timing and suggested the Caledonian orogeny was responsible through impact suturing to the east. Thus the later Early Devonian contains a major unconformity(ies), with extensive karstification likely (Okulitch *et al.* 1986). The Middle and Upper Devonian sediments in the Cornwallis Fold Belt are dominantly clastic and coarsen upwards (Kerr 1977; Embry 1988), recording the onset of the Ellesmerian orogeny. This orogeny is recorded as a compressional regime of folding and thrust faulting during latest Devonian to mid-Mississippian time (Thorsteinsson 1970, 1980, 1988) leading to dominantly east–west-trending folds, especially in the Parry Islands Fold Belt, and the onset of sedimentation into the east–west-trending Sverdrup Basin, which truncates the Cornwallis Fold Belt on the north (Fig. 1). The orogeny was followed in the Sverdrup Basin by further uplift, erosion, and clastic sedimentation related to rifting through to the start of Permian time (Beauchamp *et al.* 1989). Using a combination of ore-stage fluid-inclusion temperatures, average geothermal gradients, and palinspastic reconstructions, Kerr (1977) concluded that the Polaris mineralization was most likely Late Devonian in age. In the early Cenozoic Eurekan orogeny (Trettin 1989), major rift grabens were formed, including the one that trends north–south under Crozier Strait on the immediate west side of the Polaris deposit and which may be responsible for the small fold used for the paleomagnetic fold test (Fig. 2). The Eurekan Freemans Cove subvolcanic alkaline intrusions and vents, which have an Eocene Rb–Sr age, outcrop on the opposite side of Crozier Strait.

Experimental methods

Sampling

Three oriented blocks were collected at each of the 29 sites, with 8 in the zinc–lead ore, 12 in the surrounding dolomite envelope, and 9 in more remote limestones of the Thumb Mountain Table Formation (Table 1). The mine is located entirely in permafrost, creating two specific sampling problems. First, the walls are heavily frosted or iced, making it difficult to select and mark block samples in much of the mine. Second, blocks that may appear coherent underground may be held together by ice in fractures and so fall apart at room temperature. Yet another problem is that Polaris is only about 120 km from the north magnetic pole. Thus, the present Earth's magnetic field (PEMF) is virtually vertical, so a magnetic compass is useless for block orientation. Therefore, the blocks were oriented using 1:200 scale mine plans for topographic control, so their declinations are slightly less accurate than normal for most paleomagnetic studies. From three to seven specimens were drilled from each of the 87 blocks in the laboratory, yielding a total of about 450 specimens. The specimens were described lithologically and weighed to obtain an estimate of their sulphide content.

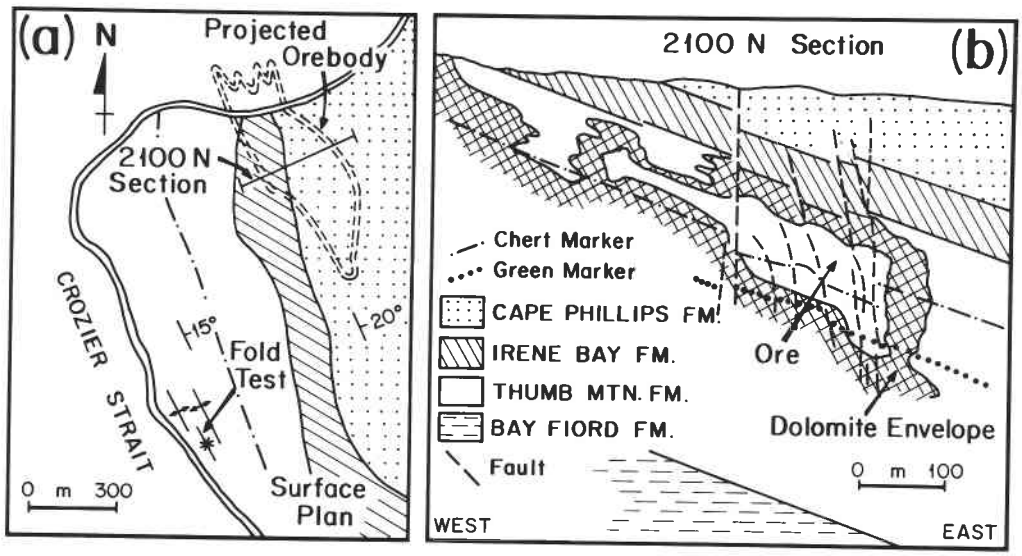


FIG. 2. Geology of the Polaris mine: (*a*) in plan and (*b*) section after Randell and Anderson (1990).

Natural remanent magnetization

The specimens were stored in a steel-shielded room with an ambient magnetic field of ≤ 100 nT for about 2 weeks prior to measurements to allow their viscous remanent magnetization (VRM) components to decay, and all subsequent measurements were done inside the room. The natural remanent magnetization (NRM) of each specimen was measured using an automated two-axis CTF DRM-420 cryogenic magnetometer with a sensitivity of about 4×10^{-6} A/m. The NRM directions were generally southeasterly with moderately steep to nearly horizontal down directions. For each site the three or four specimens with the most intense NRM were chosen from each of the three blocks to give reduced collection of 286 specimens with 9–11 specimens from each site (Table 1). For this reduced collection, the mean NRM intensity is $(1.2 \pm 0.7) \times 10^{-4}$ A/m, with a range from low 10^{-5} to mid 10^{-4} A/m.

Demagnetization procedures

One typical specimen from each site was alternating-field (AF) demagnetized in 13 steps up to 120 mT using a Sapphire Instruments SI-4 unit. These AF test specimens showed a uniform behaviour whether from limestone host rocks, dolomite envelope, or ore. The initial AF steps up to 15 or 20 mT remove a near-vertical down component that apparently reflects the acquisition of a Holocene VRM by the specimens (Figs. 3, 4). Further demagnetization up to about 80 or 100 mT isolates a characteristic magnetization component that is directed to the southeast with a shallowly to moderately upward inclination. This is termed the *A* component. Finally, as the AF step intensities climb above 100 mT, the specimens start to acquire erratic anhysteretic remanence (ARM) components. The ARM effect is exacerbated because the remanence intensity of most specimens is close to the noise level of the magnetometer by 100 mT (Fig. 3*d*).

Next, one specimen from each site was thermally demagnetized in seven steps up to 550°C. Most specimens, whether limestone, dolomite, or ore, show the initial removal of the Holocene VRM up to about 300°C (Fig. 5). By 450–500°C some specimens had to be abandoned because their sulphide minerals started to oxidize severely, giving off fumes and generating magnetic minerals with a more intense remanence and erratic directions. For some of these specimens the *A* magnetization component could be identified in steps around 400°C and up to as high as 550°C (Fig. 5). Given the low rate of success, however, further thermal demagnetization was deemed to be unwarranted.

All remaining specimens were AF demagnetized in four steps that were chosen to best define the *A* magnetization component in the last 5–35% of the NRM intensity. The *A* component end-point directions were determined for most specimens using the minimum angular deviation (MAD) method of Kirschvink (1980). MAD values up to 15° were accepted for defining a specimen's magnetization vector, however, for most vectors the value was less than 5°. For a few specimens, the *A* vector direction was determined using the remagnetization circle and combined with the end-point data using the method of Bailey and Halls (1984). The specimen directions were then used to calculate the site mean remanence directions and statistics (Fisher 1953) (Table 1).

Fold test

Sites 23–26 were collected in limestone from the westerly-dipping limb of a small anticlinal fold with a wavelength of about 20 m (Fig. 2). Applying McElhinny's (1964) fold test to all 24 site mean directions shows that the ratio of the precision parameters on tilt correction (k corrected / k uncorrected = 26.2/15.9 = 1.65, Table 2) is the same as the 1.65 value for the corresponding theoretical F-statistic at the 95% confidence level. This indicates, just barely, that the remanence was acquired before folding. The more rigorous test of McFadden and Jones (1981) shows clearly that the site means from the east- and west-dipping limbs can be drawn from one population after tilt correction at the 95% confidence level but not before tilt correction. Thus the fold test is positive and again shows that the remanence was acquired before folding.

Scanning electron microscopy

Magnetic mineral extracts were prepared from five representative samples using methods described by Pan *et al.* (1990) and their morphology and chemical composition examined using a scanning electron microscope (SEM) with energy-dispersive analysis (EDA). The one limestone extract contained only blocky magnetite crystals (Fig. 6*a*). The two

TABLE 1. Site collection, demagnetization, and mean magnetization data after tectonic correction

Site	Mine grid location: North (m)	East (m)	Elevation (m)	Lithology	Strike (°)	Dip (°)	No. of specimens (E,R,D,W)	Demag. range (mT)	Mean direction: D (°)	I (°)	α_{95} (°)	k
1	2470	1060	970	D	340	27	3,3,1,3	40–80	134	−23	17	17
2	2395	1255	930	O	340	25	7,2,1,0	20–60	134	−36	5	109
3	2260	1303	925	O	340	31	9,1,0,0	25–70	132	−34	17	9
4	2110	1425	870	Do	340	21	9,0,0,1	30–70	131	−9	13	17
5	2110	1505	875	Do	340	13	10,0,0,0	25–50	172	−26	14	13
6	2080	1330	930	D	340	21	10,0,0,0	15–85	148	−45	6	69
7	2130	1295	930	Do	340	14	7,0,0,3	20–60	138	−35	14	19
8	1820	1455	880	Od	340	31	4,3,1,1	20–80	136	−28	14	20
9	2105	1510	835	Di	340	17	7,3,0,0	25–40	157	−25	14	14
10	2110	1495	835	Od	340	17	6,3,1,0	15–80	136	−37	18	9
11	2140	1490	850	O	340	24	8,0,2,0	15–80	140	−41	16	13
12	2135	1430	850	Do	340	21	10,0,0,0	20–30	151	−26	13	14
13	2175	1575	850	D	340	16	4,3,2,1	20–50	127	12	26[a]	6
14	1825	1430	850	D	340	26	7,0,2,0	20–80	151	−42	6	92
15	1850	1590	850	O	340	28	5,3,0,2	15–30	149	−25	14	16
16	2090	1465	780	O	340	19	8,0,0,2	15–30	143	82[b]	15	15
17	1915	1555	785	Dog	340	16	3,5,1,0	20–25	127	−8	10	32
18	2165	1465	760	O	340	24	4,4,0,2	20–50	125	−15	16	13
19	2050	1460	715	Dob	340	16	5,1,0,3	20–100	155	−29	14	24
20	2140	1500	820	O	340	24	7,0,0,2	15–60	179	−15	56[a]	1
21	1800	1520	820	D	340	19	10,0,0,0	15–40	163	−37	14	12
22	1590	690	1000	L	14	5	8,0,1,1	15–90	140	−42	7	57
23	1320	670	1015	L	164	25	9,0,0,1	20–80	148	−16	13	16
24	1320	670	1015	L	161	21	10,0,0,0	20–80	160	−39	10	23
25	1320	670	1015	L	151	9	6,1,0,3	40–80	170	−37	13	23
26	1320	670	1015	L	145	19	8,0,1,1	20–80	166	−30	8	55
27	1320	650	1015	Lo	4	23	9,0,1,1	20–60	146	−44	15	12
28	1440	690	1020	Lo	320	14	2,3,0,5	15–30	161	6	46[a]	4
29	1650	630	1015	Lo	259	13	7,2,0,1	30–60	29[b]	−40	7	58

NOTES: Lithology is defined by its dominant (upper case letters) and minor (lower case letters) rock type as limestone (L), dolomite (D, d) and ore (O, o), denoting also the internal sediments (i), green marker (g), and beige marker (b). Strike and dip are given using a right-hand convention. The number of specimens with used end point or remagnetization directions is denoted by E and R, respectively, or with unused deviant or too weakly magnetized directions by D and W, respectively. The demagnetization range (demag. range) shows the spectrum used to calculate specimen vectors excluding one vector in some sites derived from the 300 to 450°C temperature range. The site mean magnetization direction is given after tectonic correction for bedding tilt and for a 3°N plunge. It is defined by its declination (D), inclination (I), radius of cone of 95% confidence (α_{95}), and precision parameter (k; Fisher 1953).

[a]Site mean excluded from unit mean calculations because of dispersed specimen vectors giving an α_{95} value of more than 19°.

[b]Site mean excluded from unit mean calculations because of a radically deviant direction uncorrected for tilt.

extracts from barren dolomite also contain mostly blocky magnetite crystals except for rare marcasite crystals. The two extracts from ore samples were more varied. Most crystals were blocky magnetite (Figs. 6*b*, 6*c*). A few crystals were found of marcasite (Fig. 6*d*), sphalerite (Fig. 6*e*) and galena. One possible spheroid of magnetite with a diameter of 1.5 μm (Fig. 6*f*) was also found in the ore extracts. The magnetite crystals are rarely more than a few micrometres in size, indicating that single-domain (SD) or pseudosingle-domain (PSD) magnetite is likely to be the dominant remanence carrier in the limestone host rocks and dolomite envelope of the Polaris ore zone. In ore the remanence is likely carried by both blocky crystals of magnetite and inclusions of magnetite in the other sulphide minerals, again of SD to PSD size.

Saturation remanence

Two tests using saturation isothermal remanent magnetization (SIRM) were run. In the first test, 10 representative specimens were magnetized using a Sapphire Instruments SI-6 pulse magnetizer in 12 steps up to 900 mT and then AF demagnetized in six steps to 40 mT. All specimens behaved similarly, reaching 96–99% saturation by 300 mT, and thereby indicating that magnetite is the only significant magnetic mineral in them (Fig. 7*a*). The saturation magnetizations (J_{900}) all fall within a narrow range from 1.3 to 13.5 $\times$ 10^{-2} A/m, with most values about 5.5 $\times$ 10^{-2} A/m (Fig. 7, caption). On AF demagnetization of this SIRM, all 10 specimens demagnetize very similarly within a narrow range between the PSD and SD magnetite type curves (Fig. 7*b*). There are some subtle differences. Both limestone specimens saturate quickly, like specimen 22 in Fig. 7*a*, whereas the dolomite and ore specimens are slightly slower to saturate. On demagnetization of the SIRM, the limestone and dolomite specimens decay nearly linearly, like specimens 12 and 29 in Fig. 7*b*, whereas the ore specimens yield a slightly exponential decay like specimens 3 and 20. We speculate that SD inclusions of magnetite in the sulphides might be responsible for these subtle differences.

The second SIRM test was run on galena crystals from Polaris because single crystals carry NRM intensities of about 2 $\times$ 10^{-3} A/m, or about two orders of magnitude greater than the host limestones or dolomites. An extended search of

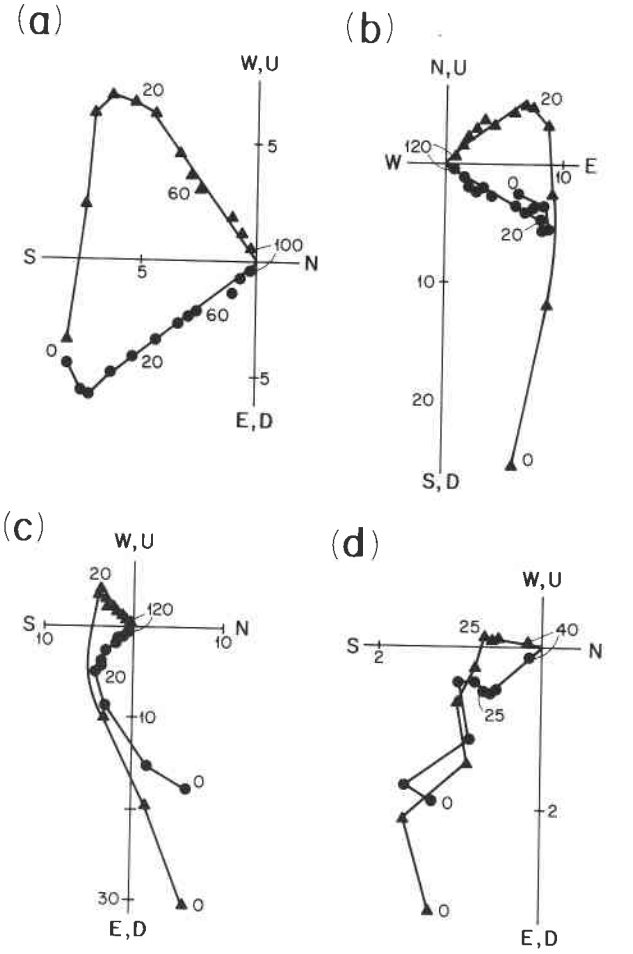


FIG. 3. Orthogonal AF demagnetization diagrams showing examples of dolomite specimens carrying the *A* component before tilt correction from sites (*a*) 6, (*b*) 14, (*c*) 19, and (*d*) 12. The axes are north (N), east (E), south (S), west (W), up (U), and down (D). The axial units are in 10^{-5} A/m. Circles and triangles denote projections onto the horizontal and vertical planes, respectively. The AF intensities are in millitesla.

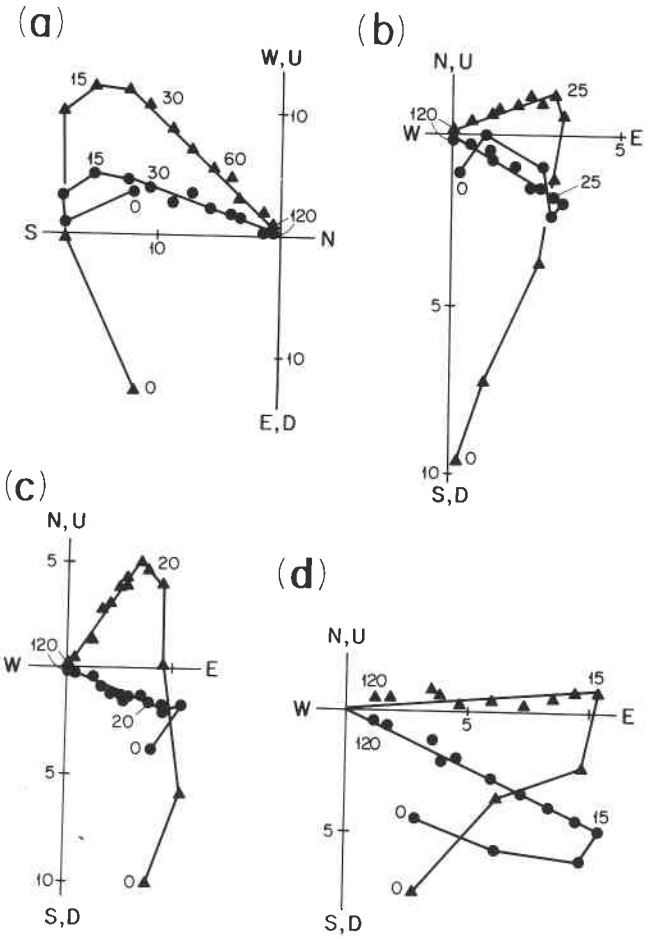


FIG. 4. Orthogonal AF demagnetization diagrams showing example specimens carrying the *A* component before tilt correction of (*a*) limestone (site 24), (*b*) ore with about 33% sulphides (site 8), (*c*) ore with about 25% sulphides (site 10), and (*d*) ore with about 25% sulphides (site 11). Conventions as in Fig. 3.

the crystals by SEM−EDA with an effective resolution of about 0.5 μm failed to reveal the source of the magnetism. Single galena crystals cannot be AF demagnetized because of the skin depth limit or thermally demagnetized because of oxidation. Therefore, a portion of one crystal was ground into a powder and dispersed in artificial specimens of plaster of Paris for AF demagnetization and of a ceramic material for thermal demagnetization. The artificial specimens contain about 6% by volume of galena and were allowed to set in the laboratory field to give them a detrital remanent magnetization (DRM). Barren plaster and ceramic specimens were also prepared for control. These artificial specimens were subjected to the same AF and thermal step demagnetization and SIRM procedures as the natural specimens except that the thermal steps went up to 600°C.

Initial AF demagnetization of the galena−plaster specimen shows simple decay of a single stable DRM magnetization component (Fig. 8*a*) that is not duplicated in the barren plaster specimen. The latter specimen has only 4% of the former's NRM intensity. The SIRM acquisition curve for the galena−plaster specimen tracks that of specimen 22 in Fig. 7*a*, and the SIRM AF demagnetization curve decays a little faster or closer to the PSD-type curve than the natural specimen curves but is otherwise very similar with a comparable shape to the ore curves (Fig. 7*b*). Thermal-step demagnetization of the galena−ceramic and ceramic blank specimens reveals simple decay of a single magnetization component with a Curie temperature of about 550°C that is indicative of titanomagnetite (Fig. 8*b*).

Discussion

The AF step demagnetization data show that the *A* magnetization component is held in high-coercivity magnetite domains of >100 mT. Thermal step demagnetization indicates that the ponent is held in magnetite domains with unblocking temperatures up to 550°C. Both AF and thermal tests show that the *A* component is held in barren limestone and dolomite and in ore specimens containing up to nearly 100% sphalerite or up to about 25% galena with additional sphalerite. The SIRM tests reveal that SD to PSD magnetite is the sole significant remanence carrier in the entire range of specimens. The SEM−EDA analyses of magnetic mineral extracts indicate that the limestones and dolomites contain only small blocky magnetite crystals as remanence carriers. They also reveal that the ore extracts contain both small blocky magnetite crystals and sphalerite, galena, and marcasite crystals with apparent SD and PSD magnetite inclusions. This last observation is sup-

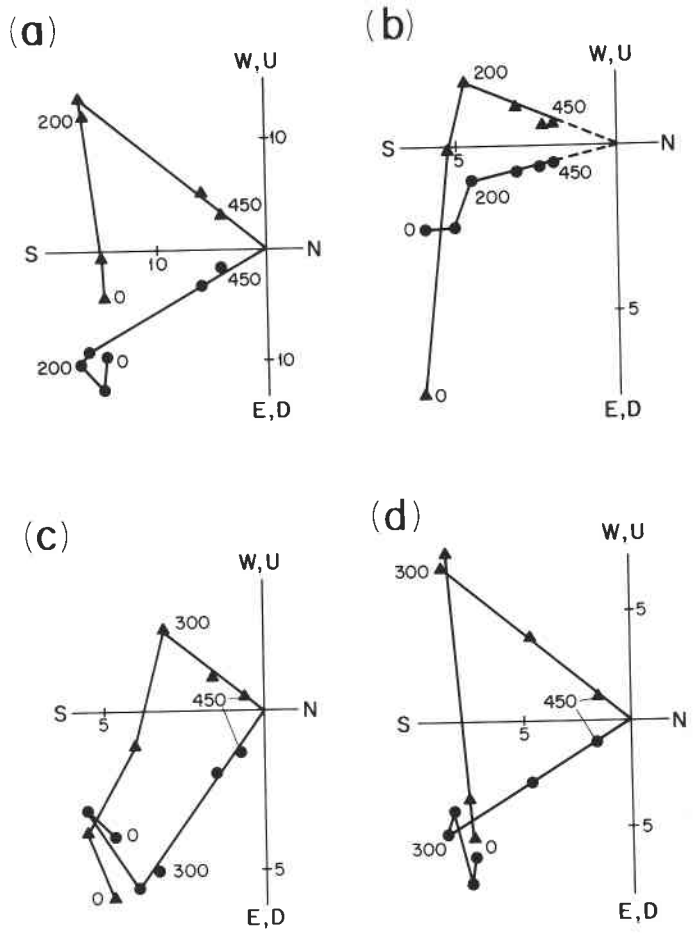


FIG. 5. Orthogonal thermal demagnetization diagrams showing example specimens carrying the *A* component before tilt correction of (*a*) limestone (site 2), (*b*) dolomite (site 19), (*c*) limestone with about 10% sulphides (site 27), and (*d*) ore with about 85% sulphides (site 20). Conventions as in Fig. 3, except that the temperatures are in degrees Celsius.

ported by the finding of a strong NRM in single galena crystals and by subsequent AF, thermal, and SIRM tests on artificial specimens containing disseminated powdered galena. This evidence argues strongly that the mineralization and magnetization events were coeval in the Polaris MVT deposit.

The site mean magnetization direction represents three geologic entities, namely the Zn−Pb ore, its dolomitic alteration envelope, and limestone host rocks. First, the means for each of these three populations after tilt correction were calculated (Table 2). Then the discrimination test of McFadden and Lowes (1981) was applied to show with >95% confidence that the ore and dolomite populations can be drawn from one population and that the limestone and combined ore-dolomite populations can be drawn from one population (Fig. 9). This result indicates that all three populations acquired their characteristic *A* magnetization component at the same time. This could happen if (*i*) the ore was essentially syngenetic, (*ii*) the primary host rock remanence was remagnetized when the ore was emplaced, or (*iii*) both ore and host rocks were remagnetized in a post-ore metamorphic event.

The Zn-Pb mineralization and its magnetization cannot be syngenetic as the discordant nature of the Polaris deposit attests (Fig. 2). Also the pole position for the *A* magnetization component of 28.6°N, 121.1°E (dp = 3.7°, dm = 6.6°, $N = 24$) lies on the Late Devonian portion of the apparent polar wander path (APWP) for the North American craton (Van der Voo 1990), truly remote from the Middle to Late Ordovician APWP position that would correspond to a syngenetic magnetization in the Thumb Mountain Formation (Fig. 10). At Polaris a mean magnetization direction of about $D = 120°$, $I = -12°$ is required to give a Middle to Late Ordovician pole which is far from the determined value of $D = 146°$, $I = -31°$.

There is also no evidence that the Polaris ore deposit and host rocks have been metamorphosed. Fluid inclusions in sphalerite indicate ore deposition temperatures in the 52−105°C range (Jowett 1975). This corresponds to a maximum burial depth at any time of about 3 km, assuming an average geothermal gradient of about 35°C/km. The conodont alteration index (CAI) about the deposit is less than 1.5, indicating that postsedimentation temperatures have not exceeded about 90°C (D. F. Sangster, unpublished data). On the other hand, the thermal demagnetization data show unblocking temperatures up to 550°C for the *A* component. This indicates that metamorphic temperatures of >400°C for about 100 Ma would have been required to reset the higher blocking temperature *A* domains (Pullaiah *et al.* 1975; Walton 1980). Thus the evidence rules out the possibility that a metamorphic event created a thermoviscous remanent overprint in the ores and host rocks (McCabe and Elmore 1990).

By elimination the evidence shows that the host limestones were remagnetized in Late Devonian time when the ore deposit was emplaced with its dolomitic halo. This supports the concept that hot brines in the 52−105°C range migrated through the limestones to remagnetize them by forming magnetite. At the same time, the brines precipitated the Polaris MVT ore and surrounding dolomite by open space filling and replacement, precipitating magnetite with a chemical remanent magnetization (CRM), either as discrete crystals or as inclusions in the ore minerals.

The fold test showed that the *A* magnetization predates folding. What is uncertain, however, is the age of the small north−south fold used in the test. The geological evidence defines three possible ages for folding: (*i*) the Early Devonian in the strongest pulse of Cornwallis Disturbance, (*ii*) the Late Devonian − Mississippian Ellesmerian orogeny; or (*iii*) the early Cenozoic Eurekan orogeny (Kerr 1977). If the ore was emplaced when the host strata were flat lying, as Randell and Anderson's (1990) evidence from internal sediments indicates, and if it is Late Devonian, then the fold is either Ellesmerian or Eurekan. Correction for the 3°N plunge is negligible and moves the pole by only 1.6° because of the nature of the dipole formula. Rotation of a thrust slice about a vertical axis, as implied by the tectonic model of Okulitch *et al.* (1986), changes the paleodeclination but not the paleoinclination, to move the pole along a small circle as shown in Fig. 11*a*. No such rotation would give either an Early Devonian or early Cenozoic pole position to correspond to rotation in pulses 2 or 3 of the Cornwallis Disturbance or in the Eurekan orogeny. To obtain an Early Devonian pole at Polaris would require the mean remanent magnetization to be directed at about $D = 150°$, $I = +25°$, and an early Cenozoic pole would require

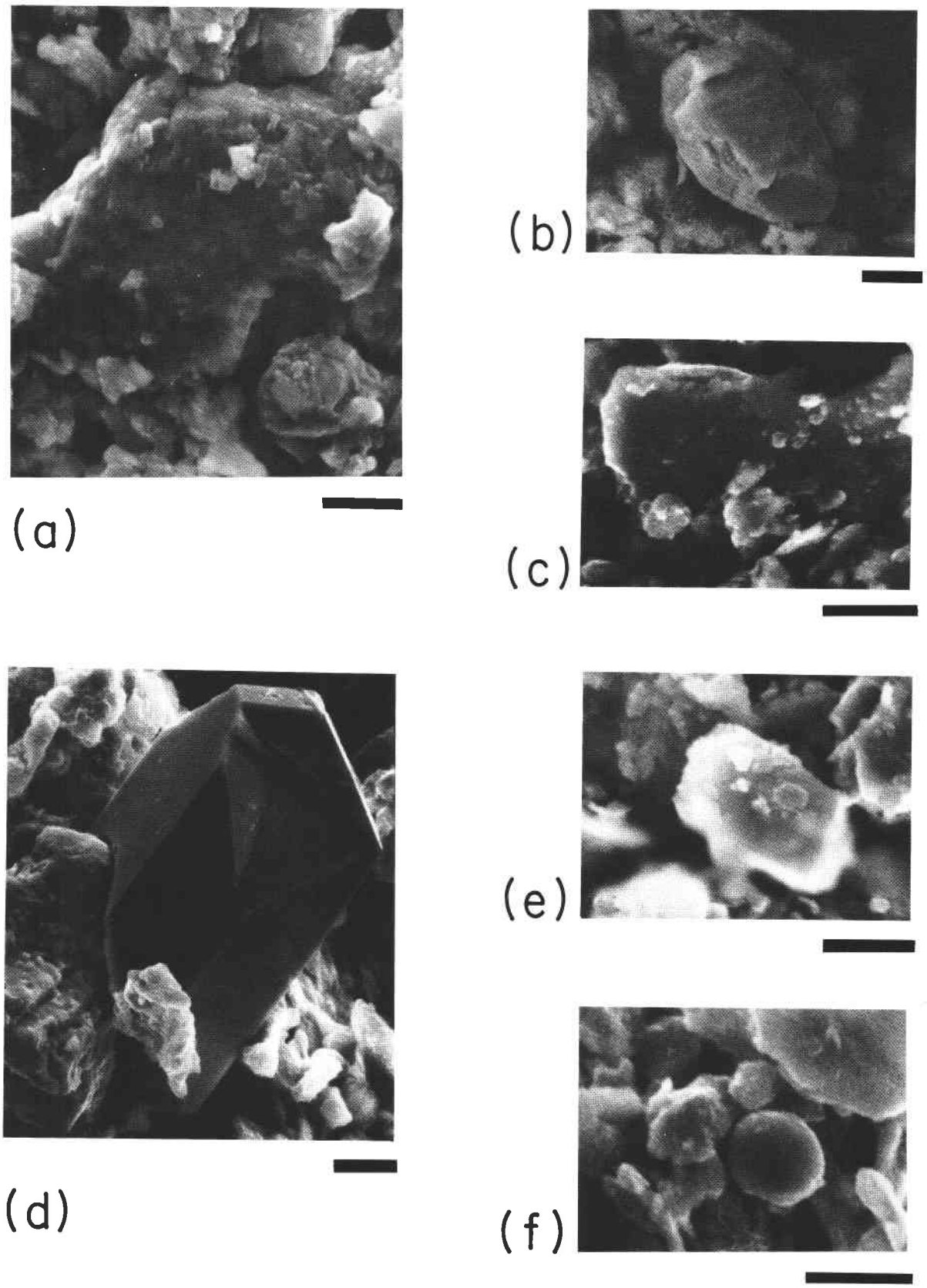


FIG. 6. Scanning electron microphotographs of minerals found in magnetic extracts, showing blocky magnetite in limestone from site 25 (*a*), in low-grade ore from site 20 (*b*), and in high-grade ore from site 3 (*c*). Other minerals in the ore magnetic mineral extract include marcasite (*d*) and sphalerite (*e*). The only possible spheroid of magnetite was found in the ore (*f*). Scale bars = 2 μm.

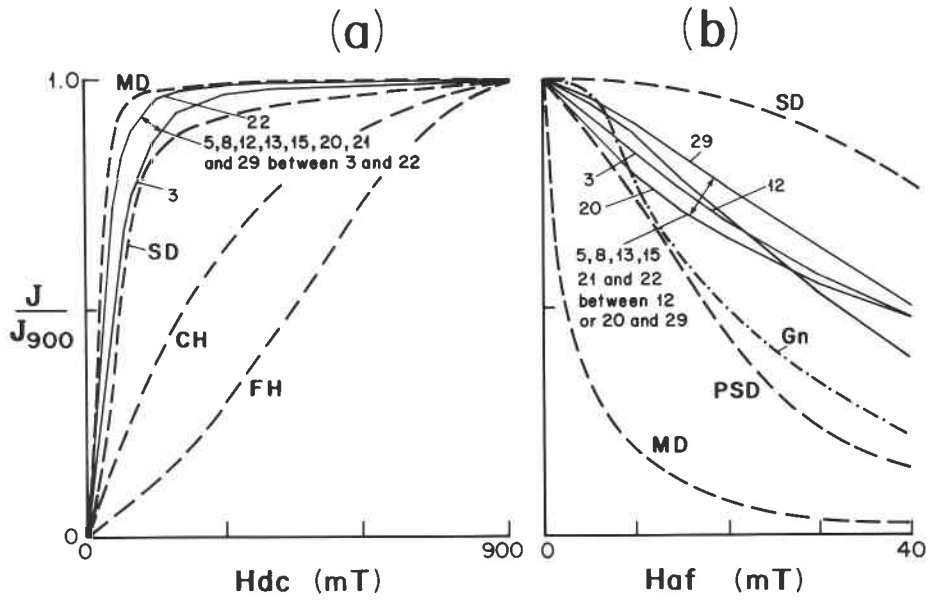


FIG. 7. Saturation isothermal remanent magnetization (SIRM) curves for example specimens showing the (*a*) acquisition of SIRM, and (*b*) demagnetization of SIRM. J/J_{900} is the ratio of the measured remanence intensity to the SIRM intensity at 900 mT, and H is the applied field in millitesla. The type curves are for single-domain (SD), pseudosingle-domain (PSD), and multidomain (MD) magnetite and for fine-grained (FH) and coarse-grained (CH) hematite from Dunlop (1971, 1972, 1973, 1981). The saturation intensities for specimens 3, 5, 8, 12, 13, 15, 20, 21, 22, and 29 are 6.0, 5.4, 4.6, 1.3, 4.7, 1.7, 13.5, 5.3, 5.8, and 6.5 × 10^{-2} A/m, respectively.

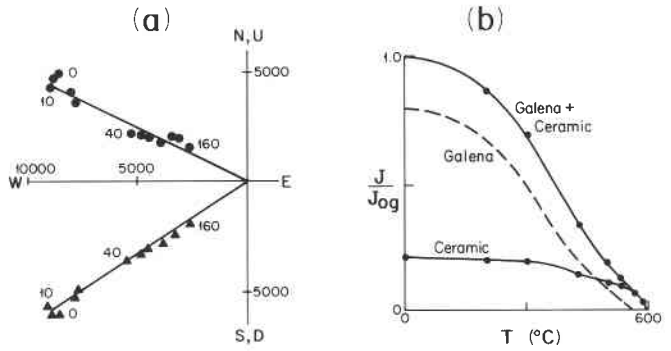


FIG. 8. (*a*) Orthogonal AF demagnetization diagram for the artificial plaster of Paris specimen containing about 6% galena by volume. Conventions as in Fig. 3. Note the relatively large remanence intensity compared with normal ore samples in Fig. 4. (*b*) Decay of remanence intensity on thermal demagnetization of artificial specimens of ceramic alone and of ceramic plus about 6% galena. The galena curve was obtained by subtraction to give a Curie temperature of about 550°C. The initial remanence intensity of the ceramic plus galena specimen (J_{og}) is 4.7 × 10^{-2} A/m.

a near-vertical remanence, both of which are far from the measured value of $D = 146°$, $I = -31°$. Thus ore deposition from low-temperature brines is coeval with the onset of the Ellesmerian orogeny in Late Devonian, followed by tilting later in the orogeny because there is no evidence for major tilting during the Eurekan. However, the minor fold could be related to an offshore Eurekan normal fault associated with rifting.

If the ore was emplaced after the host strata were tilted along the present N20°W axis as Kerr (1977) argued, then the pole would move along a great circle as shown in Fig. 11*b*. This could happen if the strata were tilted in the Early Devonian

TABLE 2. Group mean remanance directions

Group	N	Mean direction D (°)	I (°)	α_{95} (°)	k
All sites (U)	24	137.6	−26.4	7.6	15.9
Limestone sites (C)	6	155.2	−35.2	11.9	32.8
Dolomite sites (C)	11	147.4	−28.4	10.0	22.0
Ore sites (C)	7	135.9	−31.1	8.3	54.4
Dolomite + ore sites (C)	18	142.9	−29.6	6.8	26.9
All sites (C)	24	145.8	−31.1	5.9	26.2

NOTES: Conventions as in Table 1. C and U, tilt-corrected and -uncorrected site mean directions, respectively.

Cornwallis Disturbance or Late Devonian Ellesmerian orogeny, and the small fold was superimposed later in the Ellesmerian (or Eurekan) or Eurekan orogenies, respectively. Using the present average dip of about 20°E as a maximum value, the pole position swings to about 24°N, 133°E. Note that this pole is no longer a good fit to the APWP but is closest to either a Late Ordovician or a Mississippian position (Fig. 11*b*). The Late Ordovician option is rejected because there is no evidence for major pre-Devonian folding. The Mississippian option requires a complicated scenario. This hypothesis would require tilting by Late Devonian, followed by ore deposition in the Mississippian, followed by the formation of the minor fold during the Eurekan orogeny as the Crozier Strait rift developed. One difficulty is that apparently clastic sphalerite grains and clasts in the internal sediments (Randell and Anderson 1990) would have to be secondary replacements of preexisting carbonate clasts. This is unlikely, however, because the existence of abundant fragments of colloform sphalerite and mineralized rock clearly indicates the presence of sulphides before

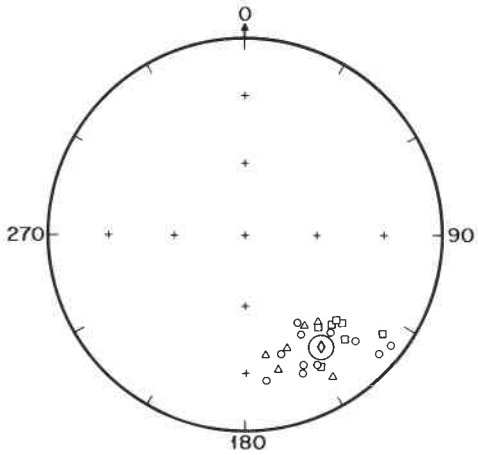


FIG. 9. Accepted mean remanence directions (Table 1) on an equal-area stereogram for the *A* magnetization component after tectonic correction for limestone (△), dolomite (○), and ore (□) sites. All are up directions. The overall mean direction (◇) is surrounded by its cone of 95% confidence (Fisher 1953).

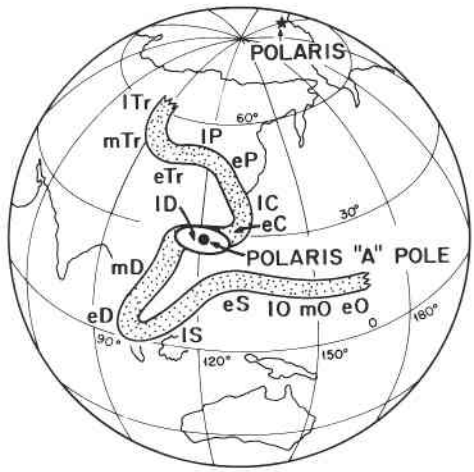


FIG. 10. Apparent polar wander path (APWP) for the North American craton from Van der Voo (1990) for the Early Ordovician to Late Triassic, with the pole for the *A* magnetization component of the Polaris ore deposit (●) surrounded by its oval of 95% confidence overlying the Late Devonian portion of the APWP. eO, mO, lO; Early, Middle and Late Ordovician; eS, lS, Early and Late Silurian; eD, mD, lD, Early, Middle, and Late Devonian; eC, lC, Early and Late Carboniferous; eP, lP, Early and Late Permian; eTr, mTr, lTr, Early, Middle, and Late Triassic.

development of the internal sediments. A second is that the fit of the ore pole to the APWP is not statistically significant at the 95% confidence level following McFadden and Lowes (1981). Note, however, that the presence of even a very small percentage of residual VRM with the *A* component would shift a Mississippian pole to the uncorrected pole position. The third problem is the complex history for the minor fold, requiring an initial 20°E dip followed by a reversal to a 20°W dip to form its west limb. Thus we do not favour the post-folding Mississippian mineralization option but concede its possibility.

Conclusions

The ore, dolomite envelope, and host limestones of the Polaris deposit all carry a prefolding *A* remanent magnetization component. It resides in PSD and SD magnetite as discrete grains and as inclusions in the MVT ore minerals. The most likely genetic timetable is that the ore deposit was formed in flat-lying Middle to Upper Ordovician host rock during the Late Devonian, coeval with the onset of the Ellesmerian orogeny, and the strata were tilted later in the orogeny. This interpretation appears to conflict with the tectonic models of Kerr (1977), Miall (1986), and Okulitch *et al.* (1986) that suggested the major folding event was Early Devonian, perhaps a distant reflection of the Caledonian orogeny, which could be expected to produce tilting by the Middle Devonian. As described by Thorsteinsson (1980), the principal crustal movements were vertical during this event and did not affect all parts of the Boothia Uplift area equally. Thus it is entirely possible that the Polaris host strata remained essentially horizontal. Note that the paleomagnetic data do not preclude a premineralization tilting of as much as 5° that was later "reinforced" to about 20° in the Ellesmerian orogeny. Alternatively, the strata in the Polaris mine area could have been flat lying at the time of mineralization because they were in a flat-lying fault block or at the crest of a broad regional anticline. It is clear from the northerly striking folds in the Disappointment Bay and younger formations that Ellesmerian deformation did occur on Little Cornwallis Island. Our results do agree, however, with

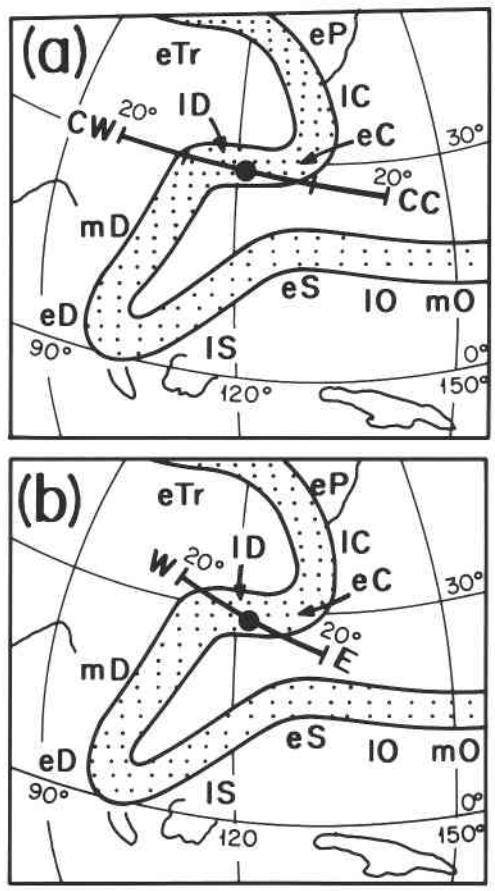


FIG. 11. Tectonic rotation of the Polaris pole in response to (*a*) clockwise (CW) or counterclockwise (CC) rotation about a vertical axis, and (*b*) rotation about a horizontal N20°W axis by 20°W to 20°E. Conventions as in Fig. 10.

Kerr's (1977) preferred Late Devonian age for the Polaris and other MVT occurrences in the Cornwallis Fold Belt. As described previously, Kerr's (1977) estimate was deduced from burial depths calculated on the basis of homogenization temperatures in sphalerite fluid inclusions. Support for at least a post-Early Devonian mineralization age is found in Kerr's Table 2, in which he lists the Stuart River and Rookery Creek MVT showings on Cornwallis Island as being hosted by the Lower Devonian Disappointment Bay Formation. This information is credited to personal communication with Cominco Limited staff (Kerr 1977, pp. 1413–1414). Recent investigation by Cominco Limited confirms the presence of mineralization in the Disappointment Bay Formation at Rookery Creek (R. Randell, personal communication, 1991). This provides independent confirmation of a post-Early Devonian lead–zinc mineralizing event in the region.

Our finding of a Late Devonian age for the Polaris Zn–Pb deposit leads to more general comment on the various models for the genesis of MVT deposits. It conflicts with the Jackson and Beales (1967) basin compaction model that should give a spectrum of Late Ordovician to Late Devonian paleomagnetic ages, corresponding to infilling of the Cornwallis Fold Belt depositional basin. It also conflicts with the Cathles and Smith (1983) model of episodic basin dewatering by basin compaction. Their model predicts MVT ore formation within about 40 Ma of host-rock deposition or a Late Ordovician – Early Silurian age. The Garven and Freeze(1984) model of gravity-driven fluid migration requires an adjacent highland for a hydraulic head to drive fluids out of a basin and into the host platform carbonates. Thus the Polaris mineralization should give a spectrum of Late Devonian to younger ages, corresponding to uplift of adjacent areas in the Ellesmerian orogeny unless mineralization is attributed to only the first flushing out of the basinal brines. The Leach and Rowan (1986) model invokes orogenic uplift and deformation as the mechanism for decanting basinal brines into foreland host carbonates. It provides an excellent fit by tying the mineralization directly to the Late Devonian maximum for the Ellesmerian Orogeny.

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